

Monte Carlo Simulation-Based Analysis of ISP Bandwidth Degradation During Peak Hours



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1. Introduction

Internet connectivity has become an essential requirement for education, business, communication, and digital services. Internet Service Providers (ISPs) advertise specific bandwidth capacities; however, users often experience reduced performance during peak usage periods. This degradation is mainly caused by congestion resulting from high simultaneous user demand.

In densely populated urban areas such as Kathmandu, bandwidth fluctuation during peak hours is common. Understanding and modeling these fluctuations is essential for evaluating ISP performance and network reliability.

Due to limited access to real ISP traffic logs, simulation-based approaches provide a practical alternative for studying network performance. This proposal presents a Monte Carlo simulation framework to model ISP bandwidth behavior under varying traffic conditions.

1.1 Background

Internet bandwidth varies significantly depending on:

- Number of concurrent users
- Available network capacity
- Time-of-day demand patterns
- Congestion effects

Peak-hour performance degradation is a common issue in broadband networks. Traditional deterministic analysis often fails to capture the randomness inherent in network traffic.

Monte Carlo simulation provides a stochastic modeling approach capable of capturing uncertain and dynamic system behavior through repeated random sampling.

This study focuses on simulating realistic ISP performance using synthetic multi-user data over a 24-hour cycle.

1.2 Objectives

This study aims to:

- To generate a synthetic multi-user ISP bandwidth dataset
- To model bandwidth variation over a 24-hour period
- To simulate network congestion using Monte Carlo methods
- To compare baseline and simulated bandwidth using mean and variance
- To analyze convergence behavior over increasing simulation iterations
- To evaluate hourly performance fluctuations

1.3 Rationale

Real-world ISP performance data is often unavailable due to:

- Privacy restrictions
- Proprietary constraints
- Limited measurement access

This creates challenges for performance analysis.

Monte Carlo simulation provides an effective alternative by:

- Modeling uncertainty
- Replicating realistic congestion behavior
- Allowing statistical evaluation of bandwidth fluctuations

The study offers a practical framework for evaluating ISP performance in resource-constrained research environments.

1.4 Assumptions

The study is based on the following assumptions:

- User arrivals follow a Poisson distribution
- Network capacity follows a normal distribution

- Peak-hour congestion occurs between 18:00 and 23:00
- Morning traffic is moderate between 06:00 and 10:00
- Off-peak hours provide higher bandwidth
- Synthetic data reasonably approximates realistic ISP behavior

1.5 Limitations

- Synthetic dataset may not capture all real-world complexities
- Assumptions on user behavior and capacity distributions
- No real ISP validation data

2. Literature Review

Network congestion analysis has been extensively studied using queueing theory and stochastic performance modeling.

Queueing theory provides the mathematical foundation for understanding communication network congestion. Kleinrock [1] established that network performance degrades non-linearly as utilization approaches capacity.

Performance evaluation methodologies developed by Jain [2] emphasize simulation-based approaches for analyzing complex computer systems under uncertainty.

Monte Carlo simulation is widely applied to stochastic system modeling. Rubinstein and Kroese [3] demonstrated its effectiveness for estimating performance metrics through repeated random sampling.

Distributed network performance and traffic management have also been studied in modern internet systems. Dilley et al. [4] describe dynamic traffic balancing mechanisms in distributed content delivery networks.

3. Materials and Methods

3.1 Data Variables

The simulation uses the following variables:

Variable	Description
user_id	Unique simulated user identifier
hour	Time of observation (0–23)
speed	Effective bandwidth (Mbps)
users	Number of active concurrent users
capacity	Available network throughput
α	Congestion sensitivity factor
efficiency	Transmission efficiency factor
noise	Random fluctuation component

3.2 Mathematical Framework / Algorithm

Bandwidth is modeled using:

$$\text{Speed} = \frac{\text{Capacity} \cdot E}{1 + \alpha U} + \epsilon$$

Where:

- Capacity = maximum ISP throughput
- U = number of concurrent users
- α = congestion sensitivity
- E = efficiency factor
- ε = random noise

Algorithm

Step 1: Generate synthetic 24-hour multi-user dataset

Step 2: Randomly select simulation hour

Step 3: Generate user load using Poisson distribution

Step 4: Generate network capacity using normal distribution

Step 5: Compute bandwidth

Step 6: Repeat over multiple iterations

Step 7: Compare results with baseline data

3.3 Error Analysis

The study evaluates simulation performance using:

Mean Comparison

Measures alignment of average bandwidth.

$$\mu = \frac{1}{n} \sum x_i$$

Variance Comparison

Measures bandwidth fluctuation.

$$\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$$

Convergence Analysis

Observes stabilization of:

- Mean estimates
- Variance estimates

over increasing simulation iterations.

Sampling Stability

Repeated sampling evaluates statistical consistency of simulation outputs.

4. Expected Outcomes

The expected outcomes include:

- A realistic synthetic ISP bandwidth model
- Accurate simulation of peak-hour congestion
- Hour-wise bandwidth performance analysis
- Convergence of Monte Carlo estimates
- Statistical validation of simulation stability

The simulation is expected to demonstrate lower bandwidth and higher variability during peak hours.

5. Work Plan of the Study

The proposed computational testing and thesis documentation layout is structured for a concise execution period, with completion projected within a timeline of one week.

References

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